

University of Engineering & Technology Lahore

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Digital Logic Design (Lab Report)

Submitted To:

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University Transport Management System

Introduction

The **University Transport Management System** is a **console-based** application developed in **C++** using **Object-Oriented Programming** concepts. The system manages university transportation services, including **vehicles, routes, students, transport passes, and administrative operations**.

The project aims to **automate** transport management tasks and reduce **manual** record keeping through file handling and data persistence.

Objectives

1. To develop a **computerized** system for managing **university transportation services** efficiently.
2. To maintain records of **students, vehicles, routes, and transport** registrations.
3. To **automate vehicle assignment, seat management, and transport pass** generation.
4. To implement **Object-Oriented Programming** concepts and **file handling** for data persistence and system reliability.

Features

- **Admin Account Management** – Create and manage admin login credentials.
- **Student Registration & Login** – Students can create accounts and securely log in.
- **Vehicle Management** – Add, view, and remove buses and vans.
- **Route Management** – Create and manage transport routes.
- **Vehicle Assignment** – Assign vehicles to specific routes.
- **Transport Registration** – Students can apply for transport services.
- **Seat Management** – Automatically tracks available and occupied seats.
- **Transport Pass Generation** – Creates transport passes with fee and fine details.
- **Report Generation** – Generates revenue and route usage reports.
- **File Handling** – Saves and loads data using text files for persistent storage.

Technologies Used

Technology	Purpose
C++	Programming Language
OOP	Software Design
File Handling	Data Storage
Visual Studio	Development Environment

OOP Concepts Implemented

Encapsulation

Data members are kept private and accessed through public member functions.

```
class Student : public User {
private:
    string department;
    bool registered;

public:
    Student();
    Student(int i, string n, string p, string d);
```

Inheritance

The User class acts as a base class.

```
class User {
protected:
    int id;
    string name;
    string password;

public:
    User();
    User(int i, string n, string p);
```

Polymorphism

Virtual functions are used in User and Vehicle classes.

```
virtual void menu() = 0;  
virtual void display();
```

Abstraction

Abstract classes:

```
virtual void displayVehicle() = 0;  
virtual void save(ofstream& out) = 0;  
virtual void load(ifstream& in) = 0;
```

```
virtual void menu() = 0;
```

Operator Overloading

Implemented in Bill class.

```
Bill operator+(Bill b);  
double total();
```

Templates

DynamicArray template class.

```
template <class T>  
class DynamicArray {  
private:  
    T* arr;  
    int size;  
};
```

Provides generic array functionality.

Classes Description

User Class

Base class for all users.

Attributes:

- id
- name
- password

Functions:

- menu()
- display()

Student Class

Derived from User.

Attributes:

- department
- registered

Functions:

- display()
- setRegistered()
- getRegistered()

Admin Class

Derived from User.

Responsibilities:

- Manage vehicles
- Manage routes
- Generate reports

Driver Class

Derived from User.

Stores driver license information.

Vehicle Class

Abstract base class.

Attributes:

- vehicleID
- numberPlate
- capacity
- occupiedSeats

Functions:

- displayVehicle()
- save()
- load()

Bus Class

Derived from Vehicle.

Additional Attribute:

- AC

Van Class

Derived from Vehicle.

Additional Attribute:

- slidingDoor

Route Class

Stores route information.

Attributes:

- routeID
- startPoint
- endPoint
- distance
- assignedVehicle

Bill Class

Handles fee calculations.

Attributes:

- fee
- fine

Functions:

- total()
- displayBill()

TransportPass Class

Represents transport registration.

Functions:

- approve()
- cancel()
- displayPass()

Report Class

Generates reports.

Functions:

- generateRevenueReport()
- generateRouteReport()

TransportManager Class

Central management class.

Responsibilities:

- Vehicle Management
- Route Management
- Student Management
- File Operations

File Handling

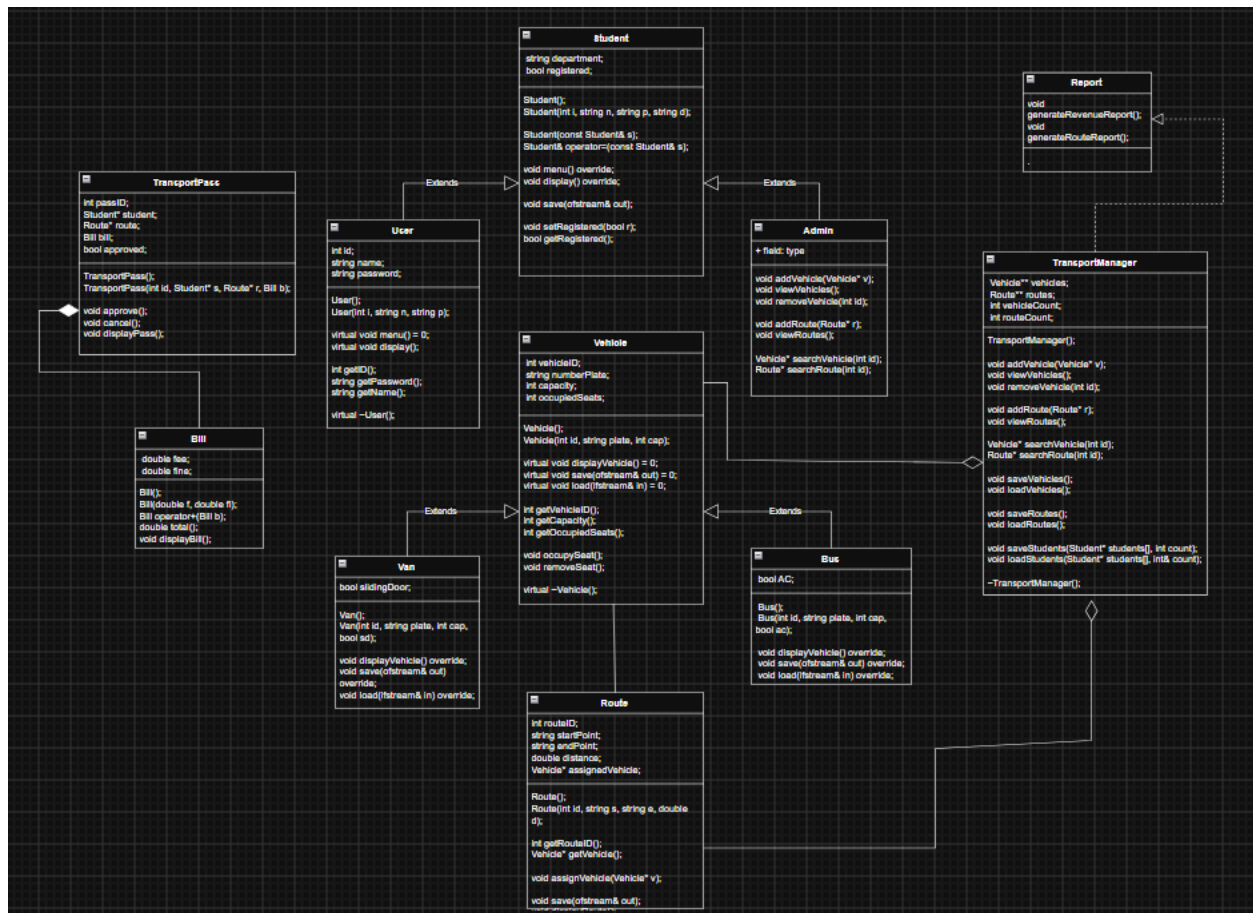
The system stores data in text files:

File Name	Purpose
vehicles.txt	Vehicle Records
routes.txt	Route Records
students.txt	Student Records
admin.txt	Admin Credentials

Benefits:

- Persistent Storage
- Data Recovery After Program Restart

UML Diagram



System Workflow

Program Start

Start Program with a menu and user choose his/her status either admin or student

```
C:\Users\Aqibr\Desktop\UETransportManagementSystem\x64\Deb
===== SYSTEM STARTED =====
===== UNIVERSITY TRANSPORT SYSTEM =====
1. Admin Login
2. Create Admin Account
3. Student Sign Up
4. Student Login
5. Exit
Enter Choice: _
```

For Admin

If user choose 1, then program requires to enter username & password followed by Admin menu

```
===== UNIVERSITY TRANSPORT SYSTEM =====
1. Admin Login
2. Create Admin Account
3. Student Sign Up
4. Student Login
5. Exit
Enter Choice: 1
Username: aqib00
Password: 124

===== ADMIN MENU =====
1. Add Vehicle
2. View Vehicles
3. Remove Vehicle
4. Add Route
5. Assign Vehicle to Route
6. View Routes
7. Reports
8. Save Data
9. Logout
Enter:
```

For Student

If the user chooses 2 and 3 then the program requires name, username, password, department & username, password respectively.

```
===== UNIVERSITY TRANSPORT SYSTEM =====
1. Admin Login
2. Create Admin Account
3. Student Sign Up
4. Student Login
5. Exit
Enter Choice:
3
ID: 1
Name: aqib
Password: 12456
Department: fs
Account Created
===== UNIVERSITY TRANSPORT SYSTEM =====
1. Admin Login
2. Create Admin Account
3. Student Sign Up
4. Student Login
5. Exit
Enter Choice: 4
ID: 1
Password: 12456
===== STUDENT MENU =====
1. View Routes
2. Apply Transport
3. View Status
4. Cancel Registration
5. View Profile
6. Logout
Enter:
```

Perform Operations

Admin Operations

Vehicle adding

```
1. Bus
2. Van
Choice:
1
ID:
4
Plate Number: 87879
Capacity: 50
AC (1/0): 0
Vehicle Added Successfully
```

Route adding

```
Enter: 4
Route ID: 5
Start: main
End: ksk
Distance: 7
Route Added
```

Assigning Routes to Vehicles

```
BUS ID: 1
Plate Number: 2997
Capacity: 50
Occupied: 1
AC: No
Route ID: 1
Vehicle ID: 1
Assigned Successfully
```

```
ROUTE ID: 1
Start: gajumatta
End: ksk
Distance: 16
Vehicle ID: 1
```

View Vehicles

```
BUS ID: 1  
Plate Number: 2997  
Capacity: 50  
Occupied: 2  
AC: No
```

Students Operations

View Routes

```
===== STUDENT MENU =====  
1. View Routes  
2. Apply Transport  
3. View Status  
4. Cancel Registration  
5. View Profile  
6. Logout  
Enter: 1  
ROUTE ID: 1  
Start: gajumatta  
End: ksk  
Distance: 16  
Vehicle ID: 1
```

Apply for Route

```
Route ID: 1  
Late days: 4  
PASS ID: 1  
Student: alishan  
Status: Approved  
FEE: 5000  
FINE: 0  
TOTAL: 5000  
Transport Approved
```

Cancel Registration

```
===== STUDENT MENU =====  
1. View Routes  
2. Apply Transport  
3. View Status  
4. Cancel Registration  
5. View Profile  
6. Logout  
Enter: 4  
Cancelled
```

View his/her profile

```
ID: 1  
Name: alishan  
Department: CS  
Status: Not Registered  
===== STUDENT MENU =====
```

Advantages

- User Friendly Interface
- Easy Record Management
- Reduced Manual Work
- Data Persistence
- Object-Oriented Design
- Reusable Code Structure

Limitations

- Console-based Interface
- Single Admin Support
- Basic Security
- Text File Storage Instead of Database
- Limited Reporting Features

Conclusion

The **University Transport Management System** successfully demonstrates the practical implementation of **Object-Oriented Programming** concepts such as **inheritance**, **polymorphism**, **abstraction**, **encapsulation**, **templates**, **operator overloading**, **exception handling**, and **file handling**. The system provides an **efficient solution** for managing university transportation services while maintaining **persistent records** through files.